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Stimulation of methane yield rate from food waste by aerobic pre-treatment

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ABSTRACT

Aerobic pre-treatment (AP) was applied to enhance methane yield from food waste through anaerobic digestion. Different AP durations (i.e. 2, 5 and 8 days) prior to anaerobic digestion were tested. The results indicated that AP of food waste led to no significant differences ($p > 0.05$) in methane yield potential (ca. 418 mL/g-VS). However, a suitable AP duration (5 days) increased methane yield rates (ca. 18 mL/d/g-VS; 22.0% higher than the control) by anticipating methane generation and shortening the methanogenic phase via volatile fatty acid reduction and pH increase. Although AP induced chemical oxygen demand loss to some extent (i.e. by 2.6%–9.9%) in the AP stage via aerobic degradation, the methane yield potential could be recovered by enhancing organic matter hydrolysis. Therefore, maximisation of hydrolysis should be used as a basis for determining a suitable AP duration for various types of organic matter.

1. Introduction

Anaerobic digestion (AD) is one of the most attractive methods used for the disposal of food waste (FW) in China owing to its potential for producing biogas and the fact that the use of anaerobic digesters is economical (Wu et al., 2015a,b, 2018; Kawai et al., 2014; Liu et al., 2013; Kim and Kim, 2013). According to literature, more than 70% of demonstration projects for FW disposal in China employ methane fermentation technology (Yu et al., 2018; Li et al., 2015). Therefore, the development of technology that stimulates methane yield from FW is of great significance.

In AD, organic matter is converted into methane by three specific bacterial groups: (i) fermentative bacteria, which hydrolyse polymers into acetic acid and other organic acids, along with hydrogen and carbon dioxide; (ii) hydrogen-producing bacteria, which convert propionate and higher fatty acids produced by the fermentative bacteria into acetate and hydrogen and (iii) methanogenic bacteria, which produce methane from acetate and/or hydrogen and carbon dioxide (Mshandete et al., 2005). However, when treating a particulate perishable substrate, e.g. FW, which is rich in carbohydrate and protein (Yu et al., 2018; Rafieenia et al., 2017), the maintenance of the balance between 'acid-forming' and 'methane-generating' microorganisms is difficult in traditional single-stage AD systems. This leads to the accumulation of volatile fatty acids (VFAs), acidification, disruption of the stability of the AD system and low methane yields (Wu et al., 2015b, 2018; Kawai et al., 2014). Therefore, the prevention of acidification is a key to ensuring stable and high-efficiency operation of AD systems.

Given that acidification of AD systems is caused by VFA accumulation (resulting in a reduction in pH of the fermentation digestate) and an imbalance between 'acid-forming' and 'acid-consuming' bacterial groups, multiple countermeasures have been developed. One such approach entails increasing the alkalinity/pH of the fermentation digestate (Liew et al., 2011; Zhu et al., 2010) and preventing the suppression of methanogenic bacteria caused by low pH. An alternative approach is to alleviate the 'burden' of the unit cell of methanogenic bacteria by decreasing the organic loading, reducing its perishable nature by mixing it with persistent materials (e.g. lignocellulosic material) and increasing the inoculum donation (i.e. methanogenic bacteria) (Wu et al., 2018; Ebner et al., 2014; Lű et al., 2012; Wang et al., 2012). However, these countermeasures may, to some extent, decrease volumetric efficiency and increase the operating costs of AD systems (Jiang et al., 2012). A further strategy in addition to a robust improvement in the methanogenic bacteria population by regulating the growth environment (addition of a specific metal/ion and redox potential

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adjustment) is directing the acidogenic metabolites towards ethanol instead of VFAs; this is achieved by adding yeast to the FW during the pre-treatment stage (Yu et al., 2018; Zhao et al., 2016; Wu et al., 2015b). Nevertheless, the mechanisms and field applicability of this technology require further study prior to its wide implementation.

To maintain the balance between hydrolytic and methanogenic reactions and reduce the retardation of methanogenic bacteria growth caused by the accumulation of VFAs, the strategy of aerobic pre-treatment (AP) of the organic matter prior to AD has been developed in recent years. In this AD scenario, organic acids generated from the organic matter hydrolysis process can be quickly decomposed. This increases the pH of the fermentative substrate, subsequently improving the start-up stability of the AD system and quickly establishing the methanogenic phase (Rafieenia et al., 2017; Xu et al., 2014, 2016). Moreover, organic matter hydrolysis, which is considered as the rate-limiting step in AD, can be promoted under aerobic conditions, likely because of the greater production of enzymes (Rafieenia et al., 2017; Peces et al., 2016; Cossu et al., 2016; Charles et al., 2009). However, inappropriate AP strategies might reduce the methane production potential owing to the loss of organic matter (Xu et al., 2014, 2016; Cossu et al., 2016; Gerassimidou et al., 2013; Brummeler and Koster, 1990). To maximise biogas recovery from organic matter, numerous studies have optimised AP conditions such as aeration rates (e.g. 0.7–32.0 NL/d/kgTS; Cossu et al., 2016), frequencies (e.g. continuous/intermittent aeration and gradually reduced aeration; Xu et al., 2016; Cossu et al., 2016; Sang et al., 2009), duration (varying from hours to months; Cossu et al., 2016; Gerassimidou et al., 2013; Mshandete et al., 2005; Zach et al., 2000) and temperature (e.g. mesophilic and thermophilic; Peces et al., 2016; Jang et al., 2014; Hasegawa et al., 2000). The influence of AP on biogas production varies according to the characteristics of the organic matter with both positive and negative impacts documented in the studies mentioned above. Formulation of the AP parameters based on MSW properties is difficult because of the heterogeneous nature of MSW. In addition, although indexes of pH, total VFAs (TVFA) and oxygen consumption have been used to determine the aeration duration (Xu et al., 2016; Cossu et al., 2016), finding a suitable specific chemical parameter that is applicable and capable of guaranteeing optimal conditions for mathanogenic bacteria is still a challenge.

An in-depth understanding of the influence of AP on organic matter fermentation processes, including both hydrolytic and methanogenic processes, is fundamentally important for improving the efficiency of AP. This study aims to evaluate the effects of AP duration on the methane yield rate of FW by examining the characteristic parameters associated with methane generation (i.e. methane yield rate, hydrolysis degree, cumulative volume produced, lag time and chemical oxygen demand (COD) reduction). This analysis will be useful for identifying the optimum AP strategy for stimulating methane production. The results thus obtained may lay the foundation for the optimal design of AP–AD systems and provide a scientific basis for further study.

2. Material and methods

2.1. Substrate and inoculum

The FW used in this study was collected from the cafeteria of a dining hall at the University of Science and Technology Beijing, China. The constituents of the FW were highly irregular, comprising mainly grain, vegetables and small amounts of meat and fish. Polyvinyl resins, egg shells and animal bones were removed before the FW was crushed in a blender (< 5-mm particle size). The FW slurry was stored at −20 °C to prevent deterioration. Anaerobic digester sludge for use as seed sludge was obtained from sedimentation tanks in the Xiaojiahe sewage treatment plant. The anaerobic digester sludge was subjected to acclimatisation for two months before the experiments. The accommodation methodology can be found elsewhere (Wu et al., 2015). The chemical characteristics of the fermentation substrate and inoculum are shown in Table 1.

Table 1
Physicochemical characteristics of food waste and inoculum used in this study.

Parameters	Food waste	Inoculum (acclimated anaerobic sludge)
TS (%, w.b.)	25.7	9.5
VS (%, w.b.)	24.1	4.9
VS/TS (%)	93.8	51.6
pH (–)	5.3	7.5
C (%, d.b.)	45.8	27.6
N (%, d.b.)	2.4	2.8
C/N (–)	18.8	9.9
COD_{Cr} (g-O_2/g-VS)	1.5	1.2
Carbohydrate (%, d.b.)	46.5	58.4
Protein (%, d.b.)	13.2	12.1
Fat (%, d.b.)	17.2	2.4
Ash (%, d.b.)	10.4	17.5

TS, total solid; VS, volatile solid; w.b., a wet basis; d.b., a dry basis.

2.2. Experimental system

Screening for the optimum AP time and the effect of inoculum-to-substrate ratio (ISR) on methane yield were conducted in wide-mouthed 500-mL Erlenmeyer flasks. Each fermentation bioreactor was sealed with a gas-tight rubber stopper and the bioreactor had an outlet equipped with a sampling septum for taking biogas samples, a 1000-mL gas-tight bag for biogas collection and a discharge outlet at the bottom with a switch to collect the fermentation digestate. The study entailed two experiments.

2.2.1. Experiment 1: screening for optimum AP duration

To examine the effect of AP duration on the subsequent performance of the batch AD of FW, AP experiments were conducted over three different durations: 2, 5 and 8 days (hereafter referred to as AP-2, AP-5 and AP-8, respectively). In this experiment, 71.2 g of FW slurry were added to the 500-mL Erlenmeyer flasks under aerobic conditions (maintained by leaving the flasks open to the environment). During the AP, the flasks were agitated three times per day, and distilled water was added to each flask to compensate for water evaporation, thus maintaining the original substrate weight. The AP experiments were conducted at a temperature of 37 ± 1 °C. Immediately after the AP step (2, 5 or 8 d), the flasks were filled with 350-mL acclimatised anaerobic digester sludge, corresponding to an ISR of 1.0 g-$VS_{inoculum}$/g-VS_{FW}. Subsequently, the flasks were flushed with nitrogen gas for 10 min to create anaerobic conditions for AD. Methane fermentation was conducted at 37 ± 1 °C, and mixing was performed at 50 rpm using an air-bath rotary shaker. In addition, a control methane fermentation treatment was prepared by mixing FW slurry and acclimatised anaerobic digester sludge (ISR 1.0) without AP. Two flasks containing only the inoculum were prepared as blanks, and the methane yield of these blanks was deducted from each treatment. The biogas production of each treatment was quantified daily, and indexes of the fermentation digestate were assayed every alternate day.

2.2.2. Experiment 2: effect of ISR on methane yield from FW that had previously been aerobically pre-treated

This experiment was performed to investigate the effect of ISR on the methane yield of FW subjected to AP. In this experiment, 78.2 g of FW slurry (instead of 71.2 g FW slurry) were subjected to AP for 5 days in Erlenmeyer flasks. The flasks were then filled with 350-mL acclimatised anaerobic digester sludge (corresponding to an ISR of 0.9 g-$VS_{inoculum}$/g-VS_{FW}), and anaerobic conditions were created by flushing with nitrogen gas prior to the subsequent AD experiment. The experimental conditions of the AD were the same as those described above. All the digestion experiments were conducted in duplicate with the average of the duplicates presented herein.

2.3. Analytical methods

The total solid (TS) and volatile solid (VS) were measured by weight loss at 105 °C and 560 °C, respectively. The carbon and nitrogen contents of the solid samples were measured using a CNH analyser (J-SCIENCE LAB Co., Japan). Values of pH were recorded using a PHS-3C type digital acidometer. Both VFAs and ethanol were measured via high-performance liquid chromatography following the method reported by Yu et al. (2018). The alkalinity of liquid samples was quantified by titration using bromocresol green–methyl red as an indicator. The daily methane yields of the bottles were calculated based on the gas generation amounts (collected by the gas sampling bags) and the methane contents in the biogas samples. The methane contents were analysed via gas chromatography (GC-2010, Shimadzu International Trading Co., Ltd., Shanghai, China) equipped with a CP-Porabond Q column and a thermal conductivity detector. The daily methane yield of each treatment was converted to the corresponding volume at standard temperature (273.15 K) and pressure (1 atm).

To mathematically simulate the cumulative methane yield of each treatment, a modified Gompertz growth equation was adopted. The model is characterised as having a sigmoidal shape with a clear inflection point (Cossu et al., 2016; Sandip et al., 2012). The equation is as follows:

$$BG = BGP \cdot \exp\left\{-\exp\cdot\left[\frac{R_m \cdot e}{BGP}(\lambda - t) + 1\right]\right\} \quad (1)$$

where *BG* (mL/g-VS) is the cumulative methane yield of the treatment, *t* (d) is the cumulated time, *BGP* (mL/g-VS) is the methane yield potential of the fermentation substrate, R_m (mL/d/g-VS) is the maximal daily methane yield, λ (d) is the duration of lag time and *e* is a mathematical constant (= 2.718). This modified Gompertz equation was calibrated by minimising the mean square deviation of the values that were measured and calculated.

3. Results

3.1. Methane yield of FW

The main results of the batch AD of FW following different AP durations are shown in terms of cumulative methane yield in Fig. 1. The cumulative methane yield ranged between 357.5 and 442.8 mL/g-VS after 31 days of digestion in all treatments. The highest methane yields recorded for ISR 1.0 were 417.3–418.6 mL/g-VS for AP-5 and the control group, followed by the yields of 392.9 and 357.7 mL/g-VS for AP-2 and AP-8, respectively. The methane yield of AP-5 at ISP 0.9 was 442.8 mL/g-VS. Notably, the methane yields over the first 24 days of fermentation (the methane yield cut-off point for AP-5 and AP-8) for AP-5 (ISR 1.0 and 0.9) were significantly ($p < 0.05$) higher than those of AP-2, AP-8 and the control group. The average methane yield rate of AP-5 (ISR 1.0 and 0.9) ranged from 17.3 to 18.5 mL/d/g-VS, which was approximately 18.6%, 18.4% and 22.0% higher than that of AP-2, AP-8 and the control group, respectively.

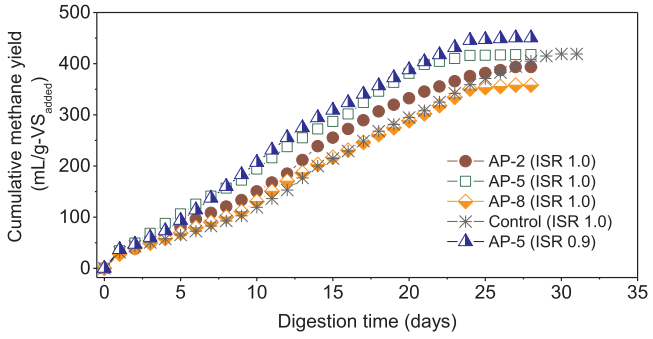


Fig. 1. Cumulative methane yields of food waste with different pre-treatment strategies.

Table 2 showed the methane yield potential, maximal daily methane yield and lag time of different treatments. Statistical indicators (R^2) were used to evaluate the reliability of the kinetic model results. The R^2 values of the treatments ranged from 0.994 to 0.998, indicating that the modified Gompertz equation is suitable to model biogas production during the test. As shown in Table 2, the fitted methane yield potentials of FW in different treatments were basically consistent to the measured methane yield potential. Additionally, the shortest methane yield lag times were 0.6 d for AP-5 (ISR 1.0), followed by 1.2, 1.6, 1.7 and 2.7 d for AP-5 (ISR 0.9), AP-2, AP-8 and the control group, respectively. The results revealed that suitable AP prior to AD could not only preserve the methane generation potential of the FW and improve the organic matter treatment capacity of the AD system but also promote the methane yield rate by shortening the methane yield lag time (compared with conventional AD systems).

3.2. Volatile fatty acid, ethanol, pH and alkalinity

Methane yield and organic matter treatment capacity depend mainly on operational parameters such as organic loading rate (OLR), pH and temperature (Wu et al., 2018; Lee et al., 2010). In particular, the evolution of VFAs plays an important role in maintaining an efficient AD as it strongly affects the pH, alkalinity and activity of methanogens (Wu et al., 2018; Zhu et al., 2010). Fig. 2 shows the changes in VFAs, ethanol, pH and alkalinity in the methanogenic reactors. In all treatments, TVFAs accumulated and reached their highest level on Day 4, decreasing thereafter. The TVFA concentrations were higher in AP-2 and the control group in the first 12 days of methanogenic fermentation. This is attributed to the low VFA consumption rates in these two treatments and is in accord with the methane yield trend observed in Fig. 1. Previous studies have suggested that a TVFA concentration > 6 g/L may result in a reduction in biogas production, whereas a value > 8 g/L may strongly inhibit AD (Karthikeyan and Visvanathan, 2013; Siegert and Banks, 2005). On this basis, AP-5 (ISR 1.0 and 0.9) and AP-8 tended to alleviate the inhibition caused by TVFAs at the beginning of fermentation (i.e. the TVFA concentration was lower than 6 g/L over the first 10 d) and consequently may have reduced the lag time, in line with the trend observed in Table 2.

In addition to the concentration of VFAs, their composition in the fermentation digestate may affect AD performance; this is a critical index reflecting the status of the AD system. Generally, there are two common pathways for the acidogenic fermentation of organic matter (i.e. propionic and butyric acid fermentation pathways), with acetic acid, propionic acid and butyric acid as the dominant VFAs (Zhu et al., 2010; Ren et al., 1995). The findings of our study showed concurrence with the expected results. It is obvious that appreciable accumulations of propionic and butyric acids were found in all treatments (Fig. 2). This observation may be accounted for as follows. First, there is a lack of propionate- and butyric-degrading syntrophic acetogenic bacteria, which are capable of transforming propionic and butyric acids to acetic acid, in the inoculum. Second, a large fraction of propionic and butyric acids was detected during the AD process, suggesting that the system became overloaded under the prevalent ISR condition (Kawai et al., 2014). A previous study demonstrated that an ISR of 1.0 is critical for the AD of FW (Wu et al., 2015). The results of this study suggest that the ISR can be further reduced to 0.9 when AP-5 is adopted, as evidenced by the methane yield of the AP-5 treatment (Fig. 1).

System pH is an important parameter for proper operation and monitoring of the AD process as it may affect acidogenic fermentation and the activity of methanogens (Wu et al., 2018; Ren et al., 1995). The pH of the fermentation digestate depends strongly on the buffering capacity, VFA presence and alkalinity of the AD system (Hernandez and Rodriguez, 2013; Zhu et al., 2010). As shown in Fig. 2, pH initially decreased with an increase in VFA production by acidogenic bacteria

Table 2
Methane production and Gompertz parameter values calculated for food waste digestion with different pre-treatment strategies.

Treatments	Experimental data			Gompertz model parameters			
	Pretreatment time (d)	Maximum daily methane production (mL/d/g-VS)	Cumulative methane production (mL/g-VS)	BGP (mL/g-VS)	R_m (mL/d/g-VS)	λ (d)	R^2 (–)
AP-2 (ISR 1.0)	2.0	26.8	392.9	467.2	19.1	1.6	0.997
AP-5 (ISR 1.0)	5.0	22.1	417.3	480.7	21.3	0.6	0.994
AP-8 (ISR 1.0)	8.0	19.8	357.7	459.2	16.4	1.7	0.996
Control (ISR 1.0)	0.0	23.4	418.6	551.0	17.3	2.7	0.997
AP-5 (ISR 0.9)	5.0	24.1	442.8	505.6	23.3	1.2	0.998

but then increased at the end of the experiment. The lowest pH of nearly 6.5 was observed in all treatments (except for AP-5 (ISR 1.0), for which the lowest pH was 6.7). Alkalinity showed a similar trend to pH. The increases in pH and alkalinity during the later stage could be owing to the consumption of VFAs by the methanogens. Notably, pH recovery (increase) in the AP-5 (ISR 1.0 and 0.9) and AP-8 treatments was faster than that in the other treatments. This is likely attributable to the relatively high pH of the fermentation digestate at the very beginning of AD (the initial pH values of the fermentation digestate were 7.9, 8.1, 8.2, 7.8 and 7.8 for AP-2, AP-5 (ISR 1.0), AP-8, the control group and AP-5 (ISR 0.9), respectively). These results indicate that suitable AP might increase the pH, and consequently the stabilisation, of the AD system via the improvement of initial digestate quality (i.e. low TVFA levels and high pH).

3.3. Organic matter hydrolysis

Given that hydrolysis of organic matter is considered the slowest system process, its stimulation may lead to more rapid AD. The hydrolysis degree (herein defined as the VS reduction rate) of each treatment at different stages is shown in Table 3. The highest total hydrolysis degree was 95.4% for AP-5 (ISP 1.0), followed by 87.6%,

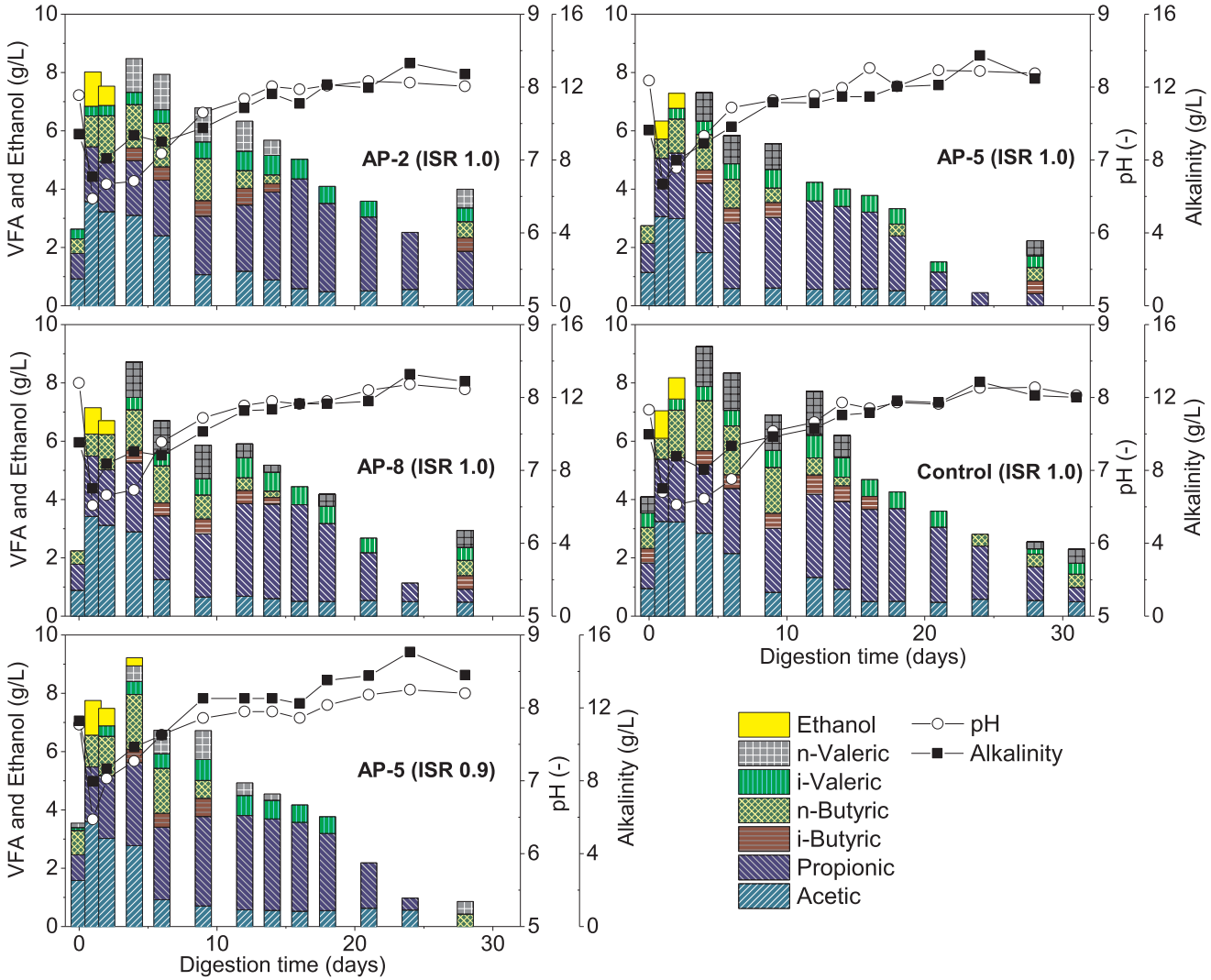


Fig. 2. Changes in ethanol concentrations, volatile fatty acid (VFA) concentrations, pH and alkalinity during digestion of food waste with different pre-treatment strategies.

Table 3
Volatile solid mass balance detected at the beginning and end of the tests on solid samples, together with the hydrolysis degree of the different treatments.

Treatments	VS of FW (g)	VS of pretreatment digestate (g)	VS of anaerobic digestion sludge (g)	VS of fermentation digestate (g)	VS of sampling[a] (g)	Hydrolysis degree in pretreatment stage (%)	Hydrolysis degree in methanogenic stage (%)	Total Hydrolysis degree (%)	Average daily hydrolysis rate[b] (%/d)
AP-2 (ISR 1.0)	17.2	15.9	17.1	16.6	2.9	7.7	78.1	85.8	3.0
AP-5 (ISR 1.0)	17.2	15.6	17.1	15.4	2.5	9.2	86.2	95.4	3.3
AP-8 (ISR 1.0)	17.2	15.4	17.1	16.6	2.7	10.4	77.2	87.6	2.7
Control (ISR 1.0)	17.2	–	17.1	16.4	3.1	–	86.3	86.3	2.9
AP-5 (ISR 0.9)	18.9	17.1	17.1	17.6	2.9	9.2	72.6	81.8	2.8

VS refers to volatile solid.
[a] VS loss due to the sampling.
[b] Defines as Total hydrolysis degree divides by total digestion time (including pretreatment time and effective methanogenic time).

86.3%, 85.8% and 81.8% for AP-8, the control group, AP-2 and AP-5 (ISP 0.9), respectively. It is noteworthy that most of the organic matter was hydrolysed during the methanogenic stage, which is consistent with previous research (Wu et al., 2018). Moreover, as expected, the increase in AP duration improved the hydrolysis degree in the pretreatment stage. The hydrolysis degree in the methanogenic stage was higher than 75% in all treatments except AP-5 (ISP 0.9). Previous research has demonstrated that an organic matter hydrolysis degree of > 75% in the methanogenic stage is crucial for maintaining the stability of AD systems (Wu et al., 2018). From this standpoint, disregarding its methane yield, the stability of the AD system in a continuous feed mode with AP-5 (ISP 0.9) still needs to be assessed. The average daily hydrolysis rate was greatest for AP-5 (ISP 1.0). Although the total hydrolysis degree of organic matter was large in AP-8, the average daily hydrolysis rate was low compared with other treatments. These results suggest that suitable AP can to some extent improve both the total hydrolysis degree and average daily hydrolysis rate.

4. Discussion

4.1. Relation between daily methane yield and process parameters

Given its high biodegradability, the irreversible acidification of the digestate (resulting from rapid hydrolysis and acidogenesis) is a major challenge to the AD of FW as it can inhibit methanogenesis and cause failure of the digestion process (Zhu et al., 2010). As an effective countermeasure, AP has been used as a means of increasing the pH of the digestate via degradation of VFAs. Moreover, AP can encourage the hydrolysis of organic matter and consequently improve the quality of the discharged fermentation digestate. The positive impact of AP on the AD of organic matter in this study is comparable with those findings listed in Table 4. However, inappropriate AP strategies (primarily an unsuitable AP duration) might lead to over-consumption of VFAs, resulting in a reduction of the methane production potential. In addition, inappropriate AP strategies might offer little advantage in terms of improving organic matter hydrolysis (as shown in Table 3). Therefore, for a specific type of organic matter, the identification of a suitable AP that allows for (i) stimulation of the methane yield rate of the organic matter and (ii) recovery of organic carbon loss via promotion of the hydrolysis degree is vital.

To ascertain the optimum AP strategy, it is necessary to unravel the correlation between daily methane yield and the various process parameters. Based on Table 5, daily methane yield has the strongest positive linear correlation with pH of the digestate. Therefore, suitable AP might increase methane yield via an increase in the pH of the digestate in the initial fermentation stage. Previous research has demonstrated that the inhibition of methanogenesis by acidification might contribute to a decrease in pH to undesirable levels, instead of VFAs accumulation, since short-chain VFAs are not toxic (Kwietniewska and Tys, 2014). Moreover, it is reported that the complete recoverable yield of methane from FW should be > 400 mL/g-VS in a stable running AD system (Kawai et al., 2014), resulting in a daily methane yield of > 13.3 mL/g-VS when the HRT of the organic matter is assumed to be 30 d. The results herein suggest that the complete recoverable yield of methane can be obtained at pH > 7.5 (Fig. 3), which is consistent with previous studies (Wu et al., 2018; Kawai et al., 2014). In addition, correlation analysis results suggested that the reduction of pH in the fermentation digestate was partially owing to the accumulation of TVFAs, acetic acid and butyric acid, indicating that the high organic loading (i.e. butyric acid accumulation) may have affected the activity of the acetic acid-consuming bacteria (i.e. methanogenesis). This was also demonstrated by the result that butyric acid had the second strongest positive linear correlation with daily methane yield (Table 5). Therefore, the rapid establishment (i.e. short lag time) of methanogenesis fermentation observed during the AP might have also been assisted by the reduction in TVFA levels at the beginning of AD (Fig. 2).

Table 4
Literature review on the influence of AP on methane yields from organic matter in batch anaerobic digestion systems.

Substrate	AP duration (d)	AP temp. (°C)	Aeration rate (m^3/kg-TS/h)	AD temp. (°C)	HRT of AD (d)	ISR (–)	OLR of AD (kg-VS/m^3/d)[b]	CH_4 yield (mL/g-VS)	References
FW	0	–	–	–	30.0	1.0	1.50	418.6	This study
	2	37	Semi-aeration[a]	37	27.0	1.0	1.66	392.9	
	5	37	Semi-aeration[a]	37	24.0	1.0	1.94	417.3	
	8	37	Semi-aeration[a]	37	24.0	1.0	1.94	357.7	
Carbohydrate rich FW	0	–	–	35	73.0	3.3	0.07	ca. 220	Rafieenia et al. (2017)
	1	NR	1.0	35	73.0	3.3	0.07	ca. 240	
Protein rich FW	0	–	–	35	73.0	3.3	0.07	241.5	
	1	NR	1.0	35	73.0	3.3	0.07	351.7	
Lipid rich FW	0	–	–	35	73.0	3.3	0.07	263.0	
	1	NR	1.0	35	73.0	3.3	0.07	240.0	
FW and office paper	0	–	–	35	535.0	NR	0.10	240.7	Gerassimidou et al. (2013)
	8	35	0.012, Continuously	35	340.0	NR	0.16	370.9	
	45	35	0.012, Continuously	35	309.0	NR	0.17	67.4	
	90	35	0.012, Continuously	35	230.0	NR	0.23	32.6	
MSW	0	–	–	39–42	305.0	NR	0.54	42.9	Cossu et al. (2016)
	55	39–42	0.0002, 12 h/d	39–42	250.0	NR	0.65	75.2	
	35	39–42	0.0002, 12 h/d	39–42	270.0	NR	0.61	101.9	
	95	39–42	0.0002, Continuously	39–42	210.0	NR	0.78	54.4	
	81	39–42	0.0002, Continuously	39–42	224.0	NR	0.73	80.5	
MSW	0	–	–	30	130.0	NR	1.93	Negligible	Xu et al. (2016)
	50	30	0.008, 8 h/d	30	80.0	NR	3.14	62.0	
	50	30	0.008, Gradually reduced	30	80.0	NR	3.14	75.0	
Primary sludge	0	–	–	37	23.0	2.0	1.81	336.0	Peces et al. (2016)
	0.5	20	Semi-aeration[a]	37	23.0	2.0	1.81	326.0	
	1	20	Semi-aeration[a]	37	23.0	2.0	1.81	340.0	
	2	20	Semi-aeration [a]	37	23.0	2.0	1.81	364.0	
	3	20	Semi-aeration[a]	37	23.0	2.0	1.81	382.0	
Sisal pulp	0	–	–	37	32.0	0.7	0.23	190.0	Mshandete et al. (2005)
	0.125	37	Semi-aeration[a]	37	32.0	0.7	0.23	190.0	
	0.25	37	Semi-aeration[a]	37	32.0	0.7	0.23	220.0	
	0.375	37	Semi-aeration [a]	37	32.0	0.7	0.23	240.0	
	0.5	37	Semi-aeration[a]	37	32.0	0.7	0.23	230.0	
	1	37	Semi-aeration[a]	37	32.0	0.7	0.23	220.0	
	2	37	Semi-aeration[a]	37	32.0	0.7	0.23	140.0	
	3	37	Semi-aeration[a]	37	32.0	0.7	0.23	120.0	

FW, food waste; HRT, hydraulic retention time; OLR, organic loading rate; NR, not reported.

[a] Semi-aerobic conditions were carried out by leaving the bottles open to the environment.

[b] OLR of AD was calculated as: original substrate concentration (kg-VS/m^3) divided by the HRT (d) of AD. The VS loss and HRT of the aerobic pre-treatment was ignored during the calculation.

Table 5
Determination coefficients between the methane yield and various investigated parameters during digestion of food waste for all pre-treatment strategies.

	COD	TVFA/ Alkalinity	TVFA	Alkalinity	pH	n-Valeric	i-Valeric	n-Butyric	i-Butyric	Propionic/ Acetic	Propionic	Acetic	Daily methane yield
Daily methane yield	−0.113	−0.352*	−0.219	0.409**	0.498**	−0.126	0.380**	−0.451**	0.008	0.440**	0.368**	−0.461**	1.000
Acetic	0.525**	0.836**	0.729**	−0.802**	−0.947**	0.274	−0.209	0.811**	0.184	−0.736**	−0.154	1.000	
Propionic	0.320*	0.238	0.398**	−0.075	−0.007	0.082	0.783**	−0.109	0.183	0.691**	1.000		
Propionic/ Acetic	−0.265	−0.539**	−0.402**	0.521**	0.634**	−0.373**	0.518**	−0.728**	−0.268	1.000			
i-Butyric	0.537**	0.595**	0.681**	−0.418**	−0.301*	0.900**	0.399**	0.543**	1.000				
n-Butyric	0.634**	0.890**	0.826**	−0.832**	−0.860**	0.614**	−0.038	1.000					
i-Valeric	0.371**	0.263	0.423**	−0.138	0.035	0.342*	1.000						
n-Valeric	0.511**	0.659**	0.713**	−0.480**	−0.408**	1.000							
pH	−0.658**	−0.903**	−0.808**	0.875**	1.000								
Alkalinity	−0.720**	−0.915**	−0.837**	1.000									
TVFA	0.762**	0.972**	1.000										
TVFA/ Alkalinity	0.775**	1.000											
COD	1.000												

TVFA refers to total volatile fatty acids.

** Correlation is significant at the 0.01 level.

* Correlation is significant at the 0.05 level.

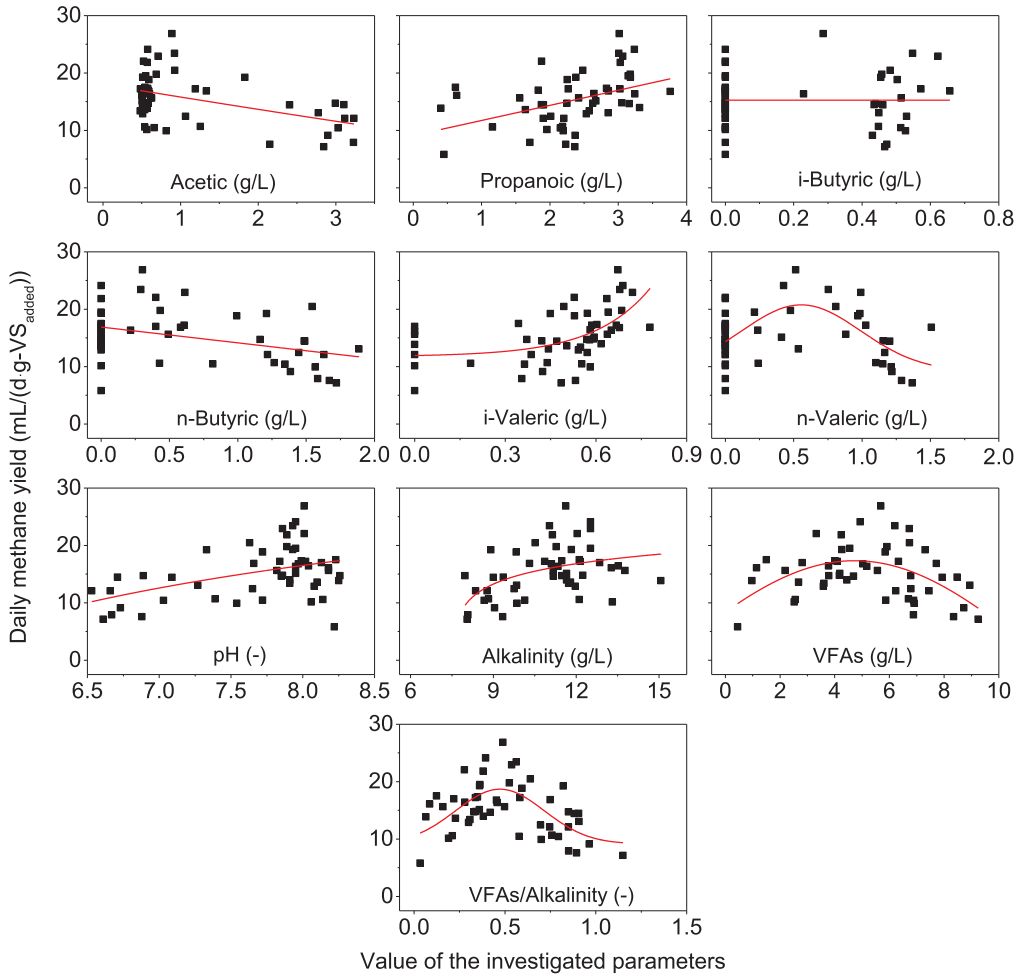


Fig. 3. Relationship between the daily methane yields and various investigated parameters under all experimental conditions. Solid lines are the fitting trend lines.

A previous study has demonstrated that the daily methane yield of AD systems is affected by the levels of propionic acid, propionic acid/acetic acid, alkalinity and TVFA/ alkalinity (Wu et al., 2015). Propionic acid is generated owing to microbial regulation of the NADH/NAD^+ balance during acetic acid generation, even in a stable AD process (Ren et al., 1995). The presence of low propionic acid levels might indicate either an insufficient digestion substrate for methanogens (inhibition of acetogenic bacteria) or that the AD system was highly functional (i.e. propionic acid was rapidly converted into acetic acid), whereas high propionic acid levels might suggest a disorder in the acetogenic and/or methanogenic processes. Therefore, relying solely on the propionic acid level to predict AD performance is not prudent. Recently, the ratio of propionic acid/acetic acid has been considered a suitable early index for predicting the inhibition or stability of AD systems; a ratio ranging from 0.03 to 0.1 is optimal for methane yield whereas a value of > 1.4 is representative of digester failure (Wu et al., 2015; Kwietniewska and Tys, 2014). However, the results of the present study differ from those obtained previously. The daily methane yield herein showed a positive linear correlation with both propionic acid and the propionic acid/acetic acid ratio. The highest propionic acid/acetic acid ratio found in this study was nearly 6.0, which is far larger than the values observed in the studies mentioned above. This may be attributable to the fact that the short-chain VFAs are not toxic and their inhibition effects are mostly contributed by the reduction in pH. The pH of the digestate was > 7.0 in all treatments; this is optimal for methanogens (i.e. pH = 7–8; Angelidaki and Sanders, 2004). Therefore, the proper control of pH in AD systems is of great importance.

The pH of the digestate in the AD system had strong positive linear correlation with alkalinity (Table 5). The results herein suggest that a daily methane yield of > 13.3 mL/g-VS (i.e. the complete recoverable yield) can be obtained when alkalinity is > 10, which is consistent with previous research (Wu et al., 2018). In contrast, alkalinity can be used to reduce the risk of over-loading AD systems via the control of the TVFA/ alkalinity ratio. It is reported that a digester can run stably at a TVFA/alkalinity ratio of < 0.4 (Gallaghan et al., 2002), 0.8 (Farquhar and Rovers, 1973) and even 1.25 (Cossu, et al., 2016); however, it has also been reported that it becomes unstable in range 0.4–0.8 (Gallaghan et al., 2002). The results of this study indicate that a satisfactory TVFA/ alkalinity ratio was in the range 0.2–0.8 (defined in terms of a daily methane yield of > 13.3 mL/g-VS) and was optimum at 0.5 (Fig. 3). Higher or lower TVFA/alkalinity ratios might produce a negative effect on methane yield owing to insufficient substrate or poor environmental conditions for methanogens (i.e. low alkalinity and pH and over-feeding).

4.2. Influence of AP strategies on COD mass balance

In the light of the above-mentioned results, the promotion of methane yield from FW can be attributed to pH/alkalinity increases due to the aerobic decomposition of VFAs. However, the mechanism relating to the effect of different AP durations on AD is still unclear. To elucidate changes in methane-forming material during AD, the COD mass balance of the different treatments was investigated. As shown in Table 6, COD loss of the substrate was larger in the AP treatment than in the control group. Moreover, a longer AP duration induced higher COD loss at a specific ISR (i.e. ISR = 1.0). Notably, extensive AP (i.e. AP-8) induced

Table 6
Chemical oxygen demand mass balance detected at the beginning and end of the test in both solid and liquid samples, together with methane yield from the food waste.

Treatments	COD of FW (g)	COD of pretreatment digestate (g)	COD of anaerobic digestion sludge (g)	COD of fermentation digestate (g)	COD of methane yield[a] (g)	COD of sampling[b] (g)	COD loss in pretreatment stage[c] (%)	COD loss in methanogenic stage[c] (%)	Total COD loss[c] (%)	Methane yield rate per gram COD[d] (mL/g-COD)	Theoretical methane yield rate per gram COD[e] (mL/g-COD)	Volumetric methane yield rate[f] (mL/mL/d)
AP-2 (ISR 1.0)	25.8	25.1	24.5	22.2	19.3	4.4	2.6	14.3	16.9	261.9	321.1	0.64
AP-5 (ISR 1.0)	25.8	24.0	24.5	19.5	20.5	3.8	6.9	17.8	24.7	278.2	364.7	0.71
AP-8 (ISR 1.0)	25.8	23.2	24.5	20.8	17.6	4.0	9.9	20.4	30.3	238.5	344.4	0.61
Control (ISR 1.0)	25.8	–	24.5	21.1	20.6	4.7	–	15.2	15.2	279.1	332.2	0.57
AP-5 (ISR 0.9)	28.4	26.6	24.5	22.1	23.9	4.3	6.3	2.6	8.9	295.2	326.5	0.81

[a] Defines as the actual methane yield divided by methane-COD conversion coefficient (i.e. 1 g COD = 350 mL methane).
[b] COD loss due to the sampling.
[c] COD loss percentage is defined as the COD loss in different stage divided by the COD of fermentation substrate (i.e. FW). The COD loss due to the sampling was excluded from the calculation of "COD loss in methanogenic stage" and "Total COD loss".
[d] Defines as the actual methane yield per gram COD of fermentation substrate (i.e. FW).
[e] The COD loss during the whole fermentation stage is converted into methane yield with the coefficient of 1 g COD = 350 mL methane. The theoretical methane yield equals to the sum of actural methane yield and methane yield of lost COD.
[f] Defines as the actual methane yield divided by the volume of the fermentation substrate and anaerobic digestion sludge and the effective fermentation time.

higher COD losses in both the pre-treatment and methanogenic stages. This may have contributed to the low methane yield observed in AP-8. Conversely, although a suitable AP strategy (i.e. AP-5 (ISR 1.0)) led to moderate COD loss, the methane production potential could be recovered by the promotion of the hydrolysis degree of the organic matter (Table 6). The methane yield rates per gram of COD were lower than the theoretical value (i.e. 1 g-COD = 350 mL methane) in all treatments owing to COD loss during AD. However, the theoretical methane yield rate per gram of COD (i.e. the COD loss during AD converted into methane yield) was close to the theoretical value and AP-5 (ISR 1.0) could even enhance the methane yield of the inoculum (defined in terms of the coefficient of 364.7 mL/g-COD). These findings are likely due to the following: (i) after suitable AP, the switch from aerobic to anaerobic conditions renders the transition from an aerobic to an anaerobic microbial population more dynamic (Cossu, et al., 2016; Jang et al., 2014) and (ii) the increased synergism among microbial species, by improving the hydrolysis of the rate-limiting step with the help of AP, might be beneficial for the methane production (Peces et al., 2016; Jang et al., 2014). In addition, although the increased OLR (i.e. low ISR) led to low COD loss (i.e. AP-5 (ISR 0.9)), it may increase the difficulty of fermentation digestate disposal (due to the high level of organic matter remaining). Regardless, the volumetric methane yield rate of the FW in AP-5 was the highest of all the treatments (Table 6), indicating the potential usefulness of the strategy.

5. Conclusions

Different AP durations were tested prior to AD. Although AP-5 showed no significant difference ($p > 0.05$) in methane yield potential compared with the control group, it stimulated the methane yield rate by anticipating methane generation and shortening the methanogenic phase resulting from VFA reduction and pH increase. Notably, although AP induced COD loss to some extent in the AP stage (2.6%–9.9%), the methane yield potential could be recovered by an increase in organic matter hydrolysis. Therefore, maximisation of the hydrolysis should be used as the basis for determining the optimum AP duration for specific organic matter types.

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